

where β is a vector of coefficients. Greene (1997, p. 874) presents models with discrete dependent variables, such as the logit model. See also Judge et al. (1985, p. 763). This may be rewritten as:

$$y = \frac{e^{X \cdot \beta}}{1 + e^{X \cdot \beta}} \times \frac{e^{-X \cdot \beta}}{e^{-X \cdot \beta}} = \frac{1}{1 + e^{-X \cdot \beta}}$$

The odds of an occurrence is:

$$\begin{aligned} odds &= \frac{y}{1-y} = \frac{\left(\frac{1}{1 + e^{-X \cdot \beta}} \right)}{1 - \frac{1}{1 + e^{-X \cdot \beta}}} \\ \Rightarrow odds &= \frac{\left(\frac{1}{1 + e^{-X \cdot \beta}} \right)}{1 - \frac{1}{1 + e^{-X \cdot \beta}}} = \frac{\left(\frac{1}{1 + e^{-X \cdot \beta}} \right)}{\left(\frac{e^{-X \cdot \beta}}{1 + e^{-X \cdot \beta}} \right)} = \frac{1}{e^{-X \cdot \beta}} = e^{X \cdot \beta} \\ &\Rightarrow \ln(odds) = X \times \beta \end{aligned}$$

The odds ratio for the control scenario ($odds_c$) versus the baseline ($odds_0$) is then:

$$odds\ ratio = \frac{odds_c}{odds_0} = \frac{\left(\frac{y_c}{1-y_c} \right)}{\left(\frac{y_0}{1-y_0} \right)} = \frac{\left(\frac{1}{e^{-X_c \cdot \beta}} \right)}{\left(\frac{1}{e^{-X_0 \cdot \beta}} \right)} = \frac{e^{X_c \cdot \beta}}{e^{X_0 \cdot \beta}}$$

The *change* in the probability of an occurrence from the baseline to the control (Δy), assuming that all of the other covariates remain constant, may be derived from this odds ratio:

$$\begin{aligned} odds\ ratio &= \frac{\frac{y_0}{1-y_0}}{\frac{y_c}{1-y_c}} = \frac{e^{X_0 \cdot \beta}}{e^{X_c \cdot \beta}} = e^{\beta \cdot \Delta X} = e^{\beta \cdot \Delta PM} \\ \frac{y_c}{1-y_c} &= \frac{y_0}{1-y_0} \times e^{-\beta \cdot \Delta PM} \end{aligned}$$

$$y_c = (1 - y_0) \times \frac{y_0}{1 - y_0} \times e^{-\beta \cdot \Delta PM}$$

$$y_c + y_c \times \frac{y_0}{1 - y_0} \times e^{-\beta \cdot \Delta PM} = \frac{y_0}{1 - y_0} \times e^{-\beta \cdot \Delta PM}$$

$$y_c \left(1 + \frac{y_0}{1 - y_0} \times e^{-\beta \cdot \Delta PM} \right) = \frac{y_0}{1 - y_0} \times e^{-\beta \cdot \Delta PM}$$

$$y_c = \frac{\frac{y_0}{1 - y_0} \times e^{-\beta \cdot \Delta PM}}{1 + \frac{y_0}{1 - y_0} \times e^{-\beta \cdot \Delta PM}} = \frac{y_0 \times e^{-\beta \cdot \Delta PM}}{1 - y_0 + y_0 \times e^{-\beta \cdot \Delta PM}} = \frac{y_0}{(1 - y_0) \times e^{\beta \cdot \Delta PM} + y_0}$$

$$\Delta y = y_0 - y_c = y_0 - \frac{y_0}{(1 - y_0) \times e^{\beta \cdot \Delta PM} + y_0}$$

$$\Delta Incidence = \Delta y \times pop = y_0 \times \left(1 - \frac{1}{(1 - y_0) \times e^{\beta \cdot \Delta PM} + y_0} \right) \times pop$$

When the coefficient (β) and its standard error (σ_β) are published (e.g., Pope et al., 1991, Table 5), then the coefficient estimates associated with the lower and upper bound may be calculated easily as follows:

$$\beta_{lowerbound} = \beta - (1.96 \times \sigma_\beta)$$

$$\beta_{upperbound} = \beta + (1.96 \times \sigma_\beta),$$

where the adjustments to the mean of plus or minus 1.96 times the standard error represent the 2.5th and 97.5th percentiles of the normal distribution, and are used to approximate a 95% confidence interval. These values can be changed to capture different lower and upper bounds.

Often the logistic regression coefficients are not published, and only the odds ratio corresponding to a specified change in PM is presented (e.g., Schwartz et al., 1994). It is easy to calculate the underlying coefficient as follows:

$$\ln(odds\ ratio) = \beta \times \Delta PM$$

$$\Rightarrow \beta = \frac{\ln(odds\ ratio)}{\Delta PM}$$

The coefficients associated with the lower and upper bound estimates of the odds ratios are calculated analogously. The underlying standard error for the coefficient (σ_β) can be approximated by:

$$\sigma_{\beta, 2.5\ percent} = \frac{\beta - \beta_{2.5\ percent}}{1.96}$$

$$\sigma_{\beta, 97.5\ percent} = \frac{\beta_{97.5\ percent} - \beta}{1.96}$$

$$\sigma_\beta \cong \frac{\sigma_{\beta, 2.5\ percent} + \sigma_{\beta, 97.5\ percent}}{2}$$

Sometimes, however, the relative risk is presented. The relative risk does not equal the odds ratio, and a different procedure should be used to estimate the underlying coefficient. Note that ESEERCO (1994, p. V-21) calculated (incorrectly) the underlying regression coefficient for Abbey et al. (1993, Table 5) by taking the logarithm of the relative risk and dividing by the change in TSP.

The relative risk (RR) is simply:

$$RR = \frac{y_0}{y_c},$$

where y_0 is the risk (i.e., probability of an occurrence) at the baseline PM exposure and y_c is the risk at the control PM exposure. When the baseline incidence rate (y_0) is given, then it is easy to solve for the control incidence rate (y_c):

$$y_c = \frac{y_0}{RR},$$

The odds ratio, may then be calculated:

$$odds\ ratio = \frac{\frac{y_0}{1 - y_0}}{\frac{y_c}{1 - y_c}}$$

Given the odds ratio, the underlying coefficient (β) may be calculated as before:

$$\beta = \frac{\ln(odds\ ratio)}{\Delta PM},$$

The odds ratio and the coefficient calculated from it are dependent on the baseline and control incidence rates. Unfortunately, it is not always clear what the baseline and control incidence rates should be. Abbey et al. (1995b, Table 2) reported that there are 117 new cases of chronic bronchitis out of a sample of 1,631, or a 7.17 percent rate. In addition, they reported the relative risk (RR = 1.81) for a new case of chronic bronchitis associated with an annual mean concentration “increment” of 45 $\mu\text{g}/\text{m}^3$ of $\text{PM}_{2.5}$ exposure.

Assuming that the baseline rate for chronic bronchitis (y_0) should be 7.17 percent, the question becomes whether the “increment” of 45 $\mu\text{g}/\text{m}^3$ should be added to or subtracted from the existing $\text{PM}_{2.5}$ concentration. If added, the control incidence rate (y_c) would be greater than the baseline rate (y_0), while subtraction would give a control rate less than the incidence rate. In effect, one might reasonably derive two estimates of the odds ratio:

$$odds\ ratio_1 = \frac{\left(\frac{y_0}{1-y_0}\right)}{\left(\frac{y_c}{1-y_c}\right)} = \frac{\left(\frac{1.81 \times 0.0717}{1-(1.81 \times 0.0717)}\right)}{\left(\frac{0.0717}{1-0.0717}\right)} = 1.931$$

$$odds\ ratio_2 = \frac{\left(\frac{y_0}{1-y_0}\right)}{\left(\frac{y_c}{1-y_c}\right)} = \frac{\left(\frac{0.0717}{1-0.0717}\right)}{\left(\frac{0.0717}{1-\frac{0.0717}{1.81}}\right)} = 1.873$$

$$\Rightarrow \beta_1 = \frac{\ln(1.931)}{45} = 0.01462$$

$$\Rightarrow \beta_2 = \frac{\ln(1.873)}{45} = 0.01394$$

An alternative is to simply assume that the relative risk (1.81) is reasonably close to the odds ratio and calculate the underlying coefficient. It is easy to show that the relative risk equals: